

1. Number of cars by Origin

Asia: 158

Europe: 123

USA: 147

2. Missing values in each column

Make: 0

Model: 0

Type: 0

Origin: 0

DriveTrain: 0

MSRP: 6

Invoice: 0

EngineSize: 0

Cylinders: 2

Horsepower: 0

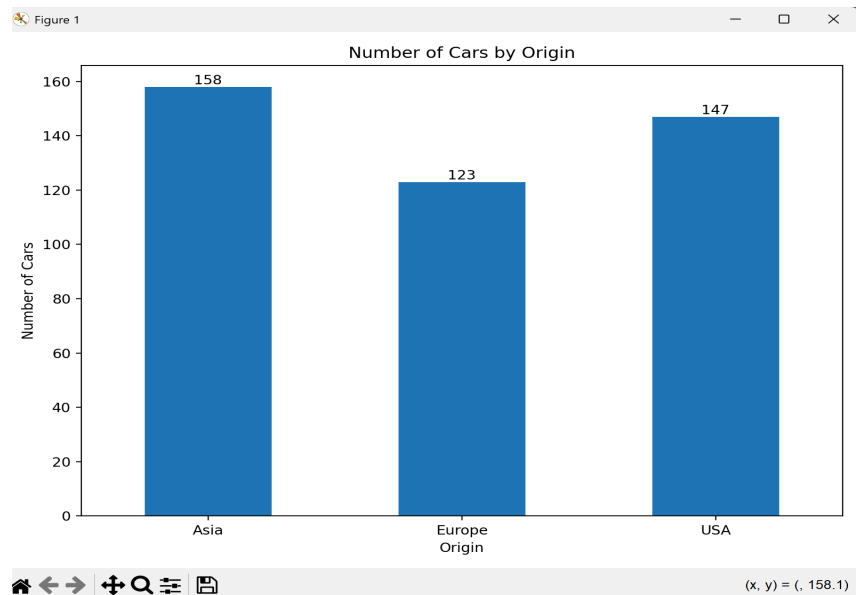
MPG_City: 0

MPG_Highway: 0

Weight: 1

Wheelbase: 0

Length: 0



3. Fuel efficiency vs Cylinders (ANOVA)

a) Statistical method

Use **One-Way ANOVA** because we are comparing the mean MPG_City across more than two groups (4, 6, and 8 cylinders).

b) Hypotheses

H₀ (Null): The mean MPG_City is the same for 4, 6, and 8 cylinder cars.

H₁ (Alternative): At least one group has a different mean MPG_City.

c) Results explanation

The F-statistic is **234.66** and the p-value is **1.18×10^{-68}** , which is much smaller than 0.05.

Since the p-value is extremely small, we **reject the null hypothesis**.

This means there is a **statistically significant difference in average MPG_City** among cars with 4, 6, and 8 cylinders. In context, fuel efficiency clearly varies depending on engine size.

4. Regression: MPG_City vs Horsepower and Weight

a) Statistical method

Use **Multiple Linear Regression** because we are predicting MPG_City using two independent variables (Horsepower and Weight).

b) Hypotheses

H₀ (Null): Horsepower and Weight have no relationship with MPG_City.

H₁ (Alternative): At least one of the variables (Horsepower or Weight) significantly affects MPG_City.

c) Results explanation

The **R² value is 0.619**, which means about **61.9% of the variation in MPG_City** is explained by Horsepower and Weight. This indicates a strong relationship.

The coefficient for **Horsepower is -0.0256**. This means that for every 1-unit increase in horsepower, MPG_City decreases by about **0.026 MPG**, holding weight constant.

The coefficient for **Weight is -0.0036**. This means that for every additional pound of weight, MPG_City decreases by about **0.0036 MPG**, holding horsepower constant.

Both variables have **p-values of 0.000**, which are less than 0.05, meaning they are statistically significant.

The model's **Mean Absolute Error is 1.80** and **RMSE is 3.23**, indicating predictions are off by about 1.8 MPG on average.

Conclusion

We **reject the null hypothesis**. Both Horsepower and Weight significantly impact MPG_City. As horsepower and weight increase, fuel efficiency decreases.

CODE:

```
import pandas as pd

import matplotlib.pyplot as plt

from scipy.stats import f_oneway

import statsmodels.api as sm

from sklearn.metrics import mean_absolute_error, mean_squared_error

import numpy as np


# 1. Import the data

cars =
pd.read_csv(r"C:\Users\AdamRay\PycharmProjects\PythonProject3\CARS.csv.csv")


print("First 5 rows of the dataset:")

print(cars.head())


print("\nColumn names:")

print(cars.columns)


# 2. Count how many cars for each Origin

origin_counts = cars.groupby("Origin").size()


print("\nNumber of cars by Origin:")

print(origin_counts)
```

```
# 3. Create bar chart for Origin counts

plt.figure(figsize=(8, 6))

ax = origin_counts.plot(kind="bar")

for i, value in enumerate(origin_counts):
    ax.text(i, value, str(value), ha="center", va="bottom")

plt.xlabel("Origin")

plt.ylabel("Number of Cars")

plt.title("Number of Cars by Origin")

plt.xticks(rotation=0)

plt.tight_layout()

plt.show()


# 4. Find missing values in each column

missing_values = cars.isnull().sum()

print("\nMissing values in each column:")

print(missing_values)


# 5. One-Way ANOVA: MPG_City across 4, 6, and 8 cylinders

mpg_4 = cars[cars["Cylinders"] == 4]["MPG_City"].dropna()

mpg_6 = cars[cars["Cylinders"] == 6]["MPG_City"].dropna()

mpg_8 = cars[cars["Cylinders"] == 8]["MPG_City"].dropna()
```

```

f_stat, p_value = f_oneway(mpg_4, mpg_6, mpg_8)

print("\nOne-Way ANOVA Results:")

print("F-statistic:", f_stat)

print("p-value:", p_value)

if p_value < 0.05:

    print("Conclusion: Reject the null hypothesis.")

    print("There is a significant difference in average MPG_City among 4, 6, and 8 cylinder cars.")

else:

    print("Conclusion: Fail to reject the null hypothesis.")

    print("There is no significant difference in average MPG_City among 4, 6, and 8 cylinder cars.")

# 6. Multiple Linear Regression: Predict MPG_City using Horsepower and Weight
regression_data = cars[["MPG_City", "Horsepower", "Weight"]].dropna()

X = regression_data[["Horsepower", "Weight"]]

y = regression_data["MPG_City"]

X = sm.add_constant(X)

model = sm.OLS(y, X).fit()

```

```
print("\nMultiple Linear Regression Results:")
```

```
print(model.summary())
```

```
# Error metrics
```

```
predictions = model.predict(X)
```

```
mae = mean_absolute_error(y, predictions)
```

```
mse = mean_squared_error(y, predictions)
```

```
rmse = np.sqrt(mse)
```

```
print("\nRegression Error Metrics:")
```

```
print("R-squared:", model.rsquared)
```

```
print("Mean Absolute Error:", mae)
```

```
print("Mean Squared Error:", mse)
```

```
print("Root Mean Squared Error:", rmse)
```